DBA5101 Analytics in Managerial Economics
Homework 1

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Question 1

(a) In a competitive market, firms are price takers and therefore set price equal to marginal cost to maximize profit.

$$P = MC = \frac{dc(q)}{dq} = 2 + 4q$$

To reverse the above equations, to express q in terms of P:

$$P = 2 + 4q \implies 4q = P - 2 \implies q = \frac{P-2}{4}$$

(b) Since there are 100 such firms in the industry, each acting as a perfect competitor, thus they share the same cost structure and marginal cost. To maximize profit, all firms will set the same price equal to their marginal cost and sell quantity would be the same.

Since each firm's supply: $q = \frac{P-2}{4}$ With 100 firms, the total market supply is:

$$Q = 100q = 100\left(\frac{P-2}{4}\right) = 25(P-2)$$

to express P in terms of Q:

$$Q = 25(P-2) \implies \frac{Q}{25} = P - 2 \implies P = 2 + \frac{Q}{25}$$

Thus, the market supply curve in inverse form is $P = 2 + \frac{Q}{25}$.

Question 2

(a) Since the total cost is $c(q) = c_v(q) + F$, the average cost would be:

$$AC(q) = \frac{c(q)}{q} = \frac{c_v(q) + F}{q} = \frac{m * q + F}{q} = m + \frac{F}{q}$$

The first order derivative of the average cost with respect to q is:

$$AC'(q) = \frac{d}{dq}\left(m + \frac{F}{q}\right) = -\frac{F}{q^2}$$

In the equation, $F > 0$ and $q^2 > 0$ (where $q \neq 0$), then the $AC'(q) < 0$ (where $q \neq 0$). Then we can conclude that the average cost is always decreasing in q.

(b) If the new company cannot differentiate its product and offers the same services, it is unlikely to survive.

We can represent the cost of the leading firm as $AC_l = m + \frac{F}{q_l}$ and the new firm as $AC_n = m + \frac{F}{q_n}$. Since $q_l > q_n$, it follows that $AC_l < AC_n$. The leading firm enjoys a significant cost advantage from economies of scale, which allows it to produce at a lower average cost compared to the new entrant.

The cost advantage allows the leading firm to use strategic pricing in order to eliminate the new entrant fairly easy:

- **Limit Pricing:** The leading firm can set a price that is between AC_l and AC_n, meaning $AC_l < p < AC_n$. In the long run, the leading company still earns a profit, while the new company incurs constant losses. This discourages entry and protects the leading firm's market position.
- **Predatory Pricing:** The leading firm can use an even more aggressive approach by temporarily setting the price below marginal cost ($MC = C_v'(q) = m$), i.e., $p < m$. Since the variable cost per unit is m, the new entrant cannot cover its variable costs and will immediately face the shutdown condition. The leading firm may incur short-term losses, but if it has accumulated sufficient profits (greater than F), it can outlast the new entrant. Once the new firm exits, the leading firm can raise prices to recoup its losses.

(c) Microsoft has high operational leverage, meaning its fixed costs are much higher than its variable costs. This creates strong economies of scale: as output increases, average cost per unit falls, giving Microsoft a cost advantage. When fixed costs F dominate the cost structure $C(q) = mq+F$, the marginal cost of serving additional customers is low, allowing Microsoft to spread fixed costs over more output and increase profitability.

(d) There are several reasons why other firms can still compete even though the industry seems like a monopoly:

- **Geopolitical factors:** Governments may support local firms for reasons such as data sovereignty and digital independence, by providing financial aid or taxes incentives. For example, in Europe, government support has enabled the growth of firms like OVHcloud and IONOS for cloud services to avoid dependence on AWS, Azure, or Google Cloud. Without such state backing, these companies would be hard to survive in a pure free market.
- **Niche differentiation:** Other companies can survive by offering specialized features or focusing on specific user needs. For example, while Microsoft Windows dominates the general desktop OS market, Apple's macOS targets users who prefer integration with Apple hardware and creative tools, and Linux distributions serve users who value open-source software, customization, or enhanced security. These alternatives address niche markets that Microsoft may not fully serve, allowing them to compete despite Microsoft's dominance.